

gov51-ps2

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<https://github.com/angelacalderonpio/gov51-ps2.git>

Gov 51 Problem Set 2

1.1 Exploring the Experiment

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(tinytex)
```

Warning: package 'tinytex' was built under R version 4.5.2

```
gay <- read_csv("data/raw/gay.csv")
```

```

Rows: 69592 Columns: 4
-- Column specification -----
Delimiter: ","
chr (1): treatment
dbl (3): study, wave, ssm

```

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
##Q1.1
```

```
nrow(gay)
```

```
[1] 69592
```

```

gay |>
  group_by(study) |>
  summarize(n = n())

```

```

# A tibble: 2 x 2
  study      n
  <dbl> <int>
1     1  59207
2     2  10385

```

```

gay|>
  group_by(treatment) |>
  summarize(n = n())

```

```

# A tibble: 5 x 2
  treatment      n
  <chr>          <int>
1 No Contact    37735
2 Recycling Script by Gay Canvasser 6514
3 Recycling Script by Straight Canvasser 6474
4 Same-Sex Marriage Script by Gay Canvasser 12452
5 Same-Sex Marriage Script by Straight Canvasser 6417

```

There are 69,592 total observations in `gay.csv`. The study has 2 unique values (1 or 2). There are 5 treatment conditions, which are: “No Contact,” “Recycling Script by Gay Canvasser,”

“Recycling Script by Straight Canvasser,” “Same-Sex Marriage Script by Gay Canvasser,” or “Same-Sex Marriage Script by Straight Canvasser.”

```
##Q1.2
```

```
gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))
```

```
# A tibble: 5 x 2
  treatment      mean_ssm
  <chr>          <dbl>
1 No Contact      3.04
2 Recycling Script by Gay Canvasser  3.13
3 Recycling Script by Straight Canvasser  3.01
4 Same-Sex Marriage Script by Gay Canvasser  3.03
5 Same-Sex Marriage Script by Straight Canvasser  3.10
```

The treatment group that shows the highest average support is Recycling Script by Gay Canvasser.

```
##Q1.3
```

```
gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(sd_ssm = sd(ssm, na.rm = TRUE))
```

```
# A tibble: 5 x 2
  treatment      sd_ssm
  <chr>          <dbl>
1 No Contact      1.68
2 Recycling Script by Gay Canvasser  1.70
3 Recycling Script by Straight Canvasser  1.68
4 Same-Sex Marriage Script by Gay Canvasser  1.67
5 Same-Sex Marriage Script by Straight Canvasser  1.68
```

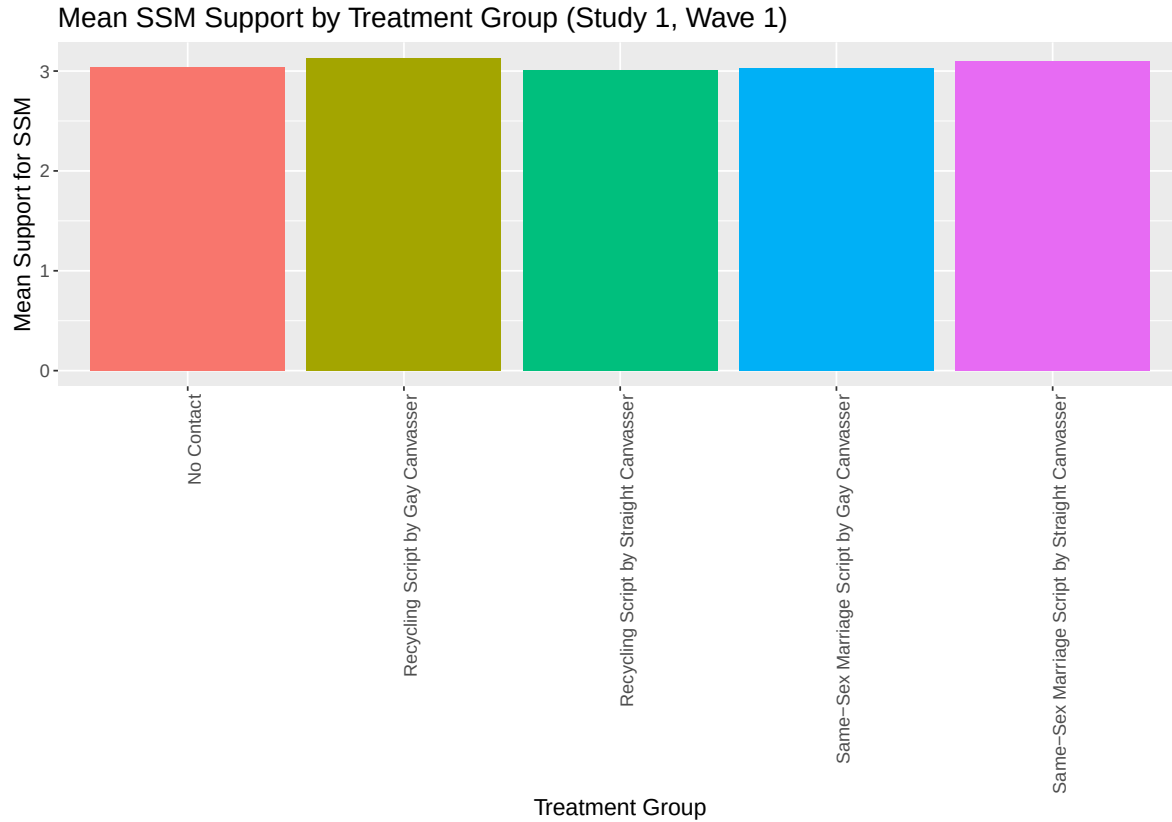
The standard deviations are roughly similar across the five treatment groups, ranging from roughly 1.66 to 1.70. This similarity is important because in a randomized experiment it suggests that the groups have comparable variability in responses as they are being drawn

from the same underlying population and randomly assigned to treatments. If the variability were very different among the groups, it could indicate that the groups were not drawn from comparable populations which would make it more difficult to attribute differences in outcome to the treatment.

```
##Q1.4

bar_plot_gay<- gay |>
  filter(study ==1, wave ==1) |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))

ggplot(bar_plot_gay, aes(x= treatment, y= mean_ssm,
                        fill = treatment)) +
  geom_col() +
  labs(x = "Treatment Group", y = "Mean Support for SSM",
       title = "Mean SSM Support by Treatment Group (Study 1, Wave 1)") +
  theme(axis.text.x = element_text(angle = 90, hjust = 1)) +
  theme(legend.position = "none") +
  theme(text = element_text(size =14))
```



```
##Q1.5
q1_data <- gay |> filter(study ==1, wave ==1)|>
filter(treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay
                        Canvasser"))

q1_data |> group_by(treatment) |>
summarize(mean_ssm = mean(ssm, na.rm = TRUE), sd_ssm = sd(ssm, na.rm = TRUE))
```

```
# A tibble: 1 x 3
  treatment mean_ssm sd_ssm
  <chr>      <dbl> <dbl>
1 No Contact    3.04  1.68
```

```
q1_data
```

```
# A tibble: 5,238 x 4
  study treatment wave  ssm
```

```

      <dbl> <chr>      <dbl> <dbl>
1         1 No Contact      1      5
2         1 No Contact      1      4
3         1 No Contact      1      5
4         1 No Contact      1      4
5         1 No Contact      1      5
6         1 No Contact      1      3
7         1 No Contact      1      2
8         1 No Contact      1      5
9         1 No Contact      1      4
10        1 No Contact      1      1
# i 5,228 more rows

```

```
(3.025195 - 3.042764) / sqrt((1.666099^2 + 1.680540^2)/2)
```

```
[1] -0.01049939
```

The standardized difference is roughly -0.01, which is much smaller than 0.25 in absolute value, suggesting that the “Same-Sex Marriage Script by Gay Canvasser” group and the “No Contact” group are well balanced at baseline. Therefore, the groups appear similar before treatment.

```

##Q1.6
q1_data2 <- gay |> filter(study ==1, wave ==2)|>
filter(treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay
                        Canvasser"))

q1_data2 |>
  group_by(treatment) |>
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))

```

```

# A tibble: 1 x 2
  treatment mean_ssm
  <chr>      <dbl>
1 No Contact    3.04

```

```
(3.139489 - 3.039615)
```

```
[1] 0.099874
```

The difference in means is 0.099874, meaning the gay canvasser group has about 0.10 higher SSM support than the no contact group in Wave 2. Because the groups were balanced at baseline, this suggests that this difference at Wave 2 reflects the average treatment effect of the canvassing treatment. The positive difference (roughly 0.10) indicates that the gay canvasser group has higher support for same-sex marriage than the no contact group. Overall, this suggests that exposure to the canvassing treatment increased support for same-sex marriage.